# Mirror/Coordinate Descent Method

*STAT 525 - Topics in Computational Statistics*

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# Overview

Mirror Descent Method

Bregman Divergence and Convergence Analysis

Coordinate Descent

Convergence Analysis

# A Different View of Gradient Descent

- Recall the gradient descent is defined as

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f(x^{(k)})$$

- Equivalently, the gradient descent can be written as

$$x^{(k+1)} = \underset{x}{\operatorname{argmin}} \left( f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) + \frac{1}{2t^{(k)}} \|x - x^{(k)}\|^2 \right).$$

  - $f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)})$ is a linear approximation of $f(x)$ at $x^{(k)}$ (the first order Taylor expansion).
  - $\|x - x^{(k)}\|^2 / 2t^{(k)}$ is the proximity term to control the discrepancy between $f(x)$ and its linear approximation (the approximation is accurate when $x$ is close to $x^{(k)}$).

# A Different View of Gradient Descent

- The quadratic proximity term $\|x - x^{(k)}\|^2/2t^{(k)}$ implicitly assume the discrepancy between $f(x)$ and its linear approximation are the same along different directions.
- If we look at the second order Taylor expansion

$$f(x) = f(x^{(k)}) + \nabla f(x^{(k)})^T(x - x^{(k)}) + \frac{1}{2}(x - x^{(k)})^T\nabla^2 f(\xi)(x - x^{(k)}),$$

  where $\xi$ is between $x$ and $x^{(k)}$, the discrepancy can be inhomogeneous.
- Sometimes, the geometry can even be non-Euclidean, e.g. the geometry in simplex.

# Quadratic Minimization

- To further illustrate this idea, we consider the quadratic minimization problem

$$\min_x \frac{1}{2} x^T Q x$$

- Recall the gradient descent method is

$$x^{(k+1)} = x^{(k)} - t^{(k)} Q x^{(k)}$$

and the convergence rate is

$$\|x^{(k)} - x^*\| \leq \left( \frac{\lambda_1(Q) - \lambda_d(Q)}{\lambda_1(Q) + \lambda_d(Q)} \right)^k \|x^{(k)} - x^*\|$$

- It can be slow when $\lambda_1(Q) \gg \lambda_d(Q)$.

# Quadratic Minimization

- The problem is that the aprroximation can be very different along different directions (think about the eigenvector of $\lambda_1(Q)$ and $\lambda_d(Q)$)

$$x^{(k+1)} = \underset{x}{\operatorname{argmin}} \left( \nabla f(x^{(k)})^T (x - x^{(k)}) + \frac{1}{2t^{(k)}} \|x - x^{(k)}\|^2 \right)$$

- What if we replace $\|x - x^{(k)}\|^2 / 2t^{(k)}$ term?

$$x^{(k+1)} = \underset{x}{\operatorname{argmin}} \left( \nabla f(x^{(k)})^T (x - x^{(k)}) + \frac{1}{2t^{(k)}} (x - x^{(k)})^T Q (x - x^{(k)}) \right)$$

implies

$$x^{(k+1)} = x^{(k)} - t^{(k)} Q^{-1} \nabla f(x^{(k)}) = (1 - t^{(k)}) x^{(k)}$$

The convergence is always very fast and has nothing to do with $Q$.

# Mirror Descent

- The idea of mirror descent is to adjust the gradient descent to fit the geormetry of the problem.
- In mirror descent, $\|x - x^{(k)}\|^2$ term is replaced by a divergence term

$$x^{(k+1)} = \operatorname*{argmin}_x \left( f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) + \frac{1}{t^{(k)}} V_w(x, x^{(k)}) \right).$$

- Here, $V_w(x, y)$ is called Bregman divergence/distance

$$V_w(x, y) = w(x) - w(y) - \nabla w(y)^T (x - y),$$

where $w$ is convex and differentiable function.

# Mirror Descent

How to choose Bregman divergence?

- Fit the the second order information very well
- Fit the geometry of constraint set
- The computation is not expensive

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# Bregman Divergence

- Recall Bregman divergence is defined as

$$V_w(x, y) = w(x) - w(y) - \nabla w(y)^T (x - y),$$

  for a strictly convex and differentiable function $w$.
- $V_w(x, y)$ can be seen as a quadratic measure defined by $w$

$$V_w(x, y) = (x - y)^T \nabla^2 w(\xi)(x - y)$$

  where $\xi$ is between $x$ and $y$.
- Choosing different $w$ yields different $V_w(x, y)$ and mirror descent update.

# Example of Bregman Divergence

- If $w$ is quadratic function

$$w(x) = \frac{1}{2} x^T Q x$$

for some positive definite matrix $Q$, then $V_w(x, y)$ is Mahalanobis-like distance

$$V_w(x, y) = \frac{1}{2}(x - y)^T Q (x - y)$$

- The mirror descent update can be written as

$$x^{(k+1)} = \operatorname*{argmin}_{x \in \Omega} \left( \langle \nabla f(x^{(k)}), x \rangle + \frac{1}{2t^{(k)}} (x - x^{(k)})^T Q (x - x^{(k)}) \right)$$

In particular, we have $x^{(k+1)} = x^{(k)} - t^{(k)} Q^{-1} \nabla f(x^{(k)})$ when $\Omega = \mathbb{R}^d$.

# Example of Bregman Divergence

- If $w$ is negative entropy

$$w(x) = \sum_{i=1}^{d} x_i \log x_i$$

defined on $\{x : \sum_i x_i = 1\}$, then $V_w(x, y)$ is Kullback–Leibler (KL) divergence

$$V_w(x, y) = \sum_{i=1}^{d} x_i \log(x_i / z_i)$$

- We leave the proof as homework.

# Example of Bregman Divergence

- If we choose the KL divergence as Bregman divergence and the simplex $\{x : \sum_i x_i = 1\}$ as constrained set, then mirror descent update is

$$x_i^{(k+1)} = \frac{x_i^{(k)} \exp(-t^{(k)} \nabla_i f(x^{(k)}))}{\sum_{j=1}^{d} x_j^{(k)} \exp(-t^{(k)} \nabla_j f(x^{(k)}))}$$

  This is also called entropic descent.
- We leave the proof as homework.
- See more examples of Bregman divergence in Dhillon, I.S. and Tropp, J.A., 2008. Matrix nearness problems with Bregman divergences. SIAM Journal on Matrix Analysis and Applications, 29(4), pp.1120-1146.

# Properties of Bregman Divergence

We will always assume $w$ is $m$-strongly convex and differentiable function on the set $\Omega$.

- Non-negativity: Since $w$ is a $m$-strongly convex function, we have

$$V_w(x, y) \geq \frac{m}{2}\|x - y\|^2$$

  Clearly, $V_w(x, y) = 0$ if and only if $x = y$.
- Convexity: $V_w(x, y)$ is a convex function in $x$
- No symmetry: $V_w(x, y) \neq V_w(y, x)$
- Linearity: For $w_1$ and $w_2$, we have $V_{w_1+aw_2}(x, y) = V_{w_1}(x, y) + aV_{w_2}(x, y)$

# Three Point Lemma

- Suppose $w$ is convex and differentiable. The following equality holds:

$$V_w(x, z) = V_w(x, y) + V_w(y, z) - (\nabla w(z) - \nabla w(y))^T (x - y).$$

- If we choose $w(x) = \|x\|^2$, then three point lemma is just law of cosine

$$\|x - z\|^2 = \|x - y\|^2 + \|y - z\|^2 - 2(z - y)^T (x - y).$$

# Three Point Lemma

*Proof:* By definition, we have

$$
\begin{aligned}
& V_w(x,y) + V_w(y,z) - V_w(x,z) \\
=& w(x) - w(y) - \nabla w(y)^T (x-y) + w(y) - w(z) - \nabla w(z)^T (y-z) \\
& - w(x) + w(z) + \nabla w(z)^T (x-z) \\
=& (\nabla w(z) - \nabla w(y))^T (x-y)
\end{aligned}
$$

# Proximal Mapping in Mirror Descent

- A proximal mapping in mirror descent method is defined as

$$\mathrm{Prox}_{\phi}(x) = \underset{z\in\Omega}{\mathrm{argmin}} \left(\phi(z) + V_w(z, x)\right).$$

- In each step, we can set $\phi^{(k)}(z) = t^{(k)}\langle \nabla f(x^{(k)}), z\rangle$, then

$$x^{(k+1)} = \mathrm{Prox}_{\phi^{(k)}}\left(x^{(k)}\right)$$

- We can show that, for any $y \in \Omega$, we have

$$\langle \nabla w(x) - \nabla w(\mathrm{Prox}_{\phi}(x)), y - \mathrm{Prox}_{\phi}(x)\rangle \leq \phi(y) - \phi(\mathrm{Prox}_{\phi}(x))$$

# Proximal Mapping in Mirror Descent

*Proof:* Recall $z^*$ is an optimal solution of optimization problem $\min_{z\in\Omega} f(z)$ if and only if

$$\langle \nabla f(z^*), y - z^* \rangle \geq 0, \qquad y \in \Omega.$$

This implies, for $y \in \Omega$ and $z^* = \text{Prox}_\phi(x)$

$$\langle \nabla \phi(z^*) + \nabla w(z^*) - \nabla w(x), y - z^* \rangle \geq 0$$

which leads to

$$\langle \nabla w(x) - \nabla w(z^*), y - z^* \rangle \leq \langle \nabla \phi(z^*), y - z^* \rangle \leq \phi(y) - \phi(z^*)$$

# Fundamental Inequality

- We now establish a fundamental inequality satisfied by the sequence generated by the mirror descent method.

$$t^{(k)}(f(x^{(k)})-f(x^*)) \leq V_w(x^*,x^{(k)})-V_w(x^*,x^{(k+1)})+\frac{{t^{(k)}}^2}{2m}\|\nabla f(x^{(k)})\|_*^2$$

- Here, we assume $w$ is $m$-strongly convex function and $\|\cdot\|_*$ is the dual norm of $\|\cdot\|$, i.e. $\|z\|_* = \sup_x\{z^T x | \|x\| \leq 1\}$.

# Fundamental Inequality

*Proof:* Since $x^{(k+1)} = \text{Prox}_{\phi^{(k)}}(x^{(k)})$ for $\phi^{(k)}(z) = t^{(k)}\langle \nabla f(x^{(k)}), z\rangle$, then

$$\langle \nabla w(x^{(k)}) - \nabla w(x^{(k+1)}), y - x^{(k+1)}\rangle \leq t^{(k)}\langle \nabla f(x^{(k)}), y - x^{(k+1)}\rangle$$

By three point lemma, we have

$$\begin{aligned} & V_w(y, x^{(k)}) - V_w(y, x^{(k+1)}) - V_w(x^{(k+1)}, x^{(k)}) \\ =& (\nabla w(x^{(k+1)}) - \nabla w(x^{(k)}))^T (y - x^{(k+1)}). \end{aligned}$$

This suggests that

$$\begin{aligned} & t^{(k)}\langle \nabla f(x^{(k)}), x^{(k)} - y\rangle \\ \leq & V_w(y, x^{(k)}) - V_w(y, x^{(k+1)}) - V_w(x^{(k+1)}, x^{(k)}) + t^{(k)}\langle \nabla f(x^{(k)}), x^{(k)} - x^{(k+1)}\rangle \end{aligned}$$

# Fundamental Inequality

*Proof:* Since $w$ is $m$-strongly convex,

$$V_w(x^{(k+1)}, x^{(k)}) \geq \frac{m}{2}\|x^{(k+1)} - x^{(k)}\|^2.$$

By Young's inequality (or Fenchel's inequality), we have

$$t^{(k)}\langle \nabla f(x^{(k)}), x^{(k)} - x^{(k+1)}\rangle \leq \frac{{t^{(k)}}^2}{2m}\|\nabla f(x^{(k)})\|_*^2 + \frac{m}{2}\|x^{(k+1)} - x^{(k)}\|^2.$$

We then have

$$t^{(k)}\langle \nabla f(x^{(k)}), x^{(k)} - y\rangle \leq V_w(y, x^{(k)}) - V_w(y, x^{(k+1)}) + \frac{{t^{(k)}}^2}{2m}\|\nabla f(x^{(k)})\|_*^2$$

We can complete the proof by setting $y = x^*$ and using the definition of convex function.

# Convergence Rate

- With the fundamental inequality, we can show that

$$\min_{0\le i\le k} f(x^{(i)}) - f^* \le \frac{V_w(x^*, x^{(0)}) + \sum_{i=0}^{k} {t^{(i)}}^2 \|\nabla f(x^{(i)})\|_*^2/2m}{\sum_{i=0}^{k} t^{(i)}}$$

- If we assume $\|\nabla f(x^{(i)})\|_* \le M$, $V_w(x^*, x^{(0)}) \le D$ and choose $t^{(k)} = O(\sqrt{2mD}/M\sqrt{k})$, then

$$\min_{0\le i\le k} f(x^{(i)}) - f^* \le O\left(\frac{M\sqrt{D}\log k}{\sqrt{mk}}\right).$$

# Convergence Rate

*Proof:* Applying fundamental inequality and taking sum yields

$$\sum_{i=0}^{k} t^{(i)}(f(x^{(i)}) - f(x^*)) \leq V_w(x^*, x^{(0)}) + \sum_{i=0}^{k} \frac{{t^{(i)}}^2}{2m} \|\nabla f(x^{(i)})\|_*^2$$

We can complete the proof by noting that

$$\sum_{i=0}^{k} t^{(i)}(\min_{0 \leq i \leq k} f(x^{(i)}) - f(x^*)) \leq \sum_{i=0}^{k} t^{(i)}(f(x^{(i)}) - f(x^*)).$$

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# Coordinate Descent

- We introduce coordinate descent method, which only search along each coordinate direction at each iteration.
  - Pick a coordinate direction $i$ from $1, \ldots, d$
  - Update $i$th coordinate

$$x_i^{(k+1)} = \underset{x_i \in \mathbb{R}}{\operatorname{argmin}} f(x_i, x_{-i}^{(k)}) \quad \text{or} \quad x_i^{(k+1)} = x_i^{(k)} - t^{(k)} \nabla_i f(x^{(k)})$$

  Here, $x_{-i}^{(k)}$ represents all other coordinates.
- This method is simple, as we only need to solve one dimensional problem.
- Coordinate descent method usually requires more steps than gradient descent, but the cost of each step is sometimes lower.

# Coordinate Descent

- There are many ways to define coordinate descent method based on different ways for updating and coordinate selecting.
- Different ways to define $x_{-i}^{(k)}$
  - Gauss-Seidel style

    $$x_{-i}^{(k)} = (x_1^{(k+1)}, \ldots, x_{i-1}^{(k+1)}, x_{i+1}^{(k)}, \ldots, x_d^{(k)})$$

    We update the coordinate if we have any. This can usually help achieve faster convergence.
  - Jacobi style

    $$x_{-i}^{(k)} = (x_1^{(k)}, \ldots, x_{i-1}^{(k)}, x_{i+1}^{(k)}, \ldots, x_d^{(k)})$$

    We hold the update until we complete all coordinates. This can be used in distributed optimization.

# Coordinate Descent

- Different ways to select coordinates
  - Cyclic: we basically run all the coordinates in the cyclic order $1, 2, 3, \ldots, d, 1, 2, 3, \ldots$
  - Random sampling: in each iteration, we randomly pick one (with some distribution)
  - Gauss-Southwell: we always pick the coordinate with largest derivative
  - Random Permutation: we run all the coordinates in the cyclic order on a permuted one.

# Example of Coordinate Descent

- Lasso problem is

$$\min_{\beta} \frac{1}{2}\|Y - X\beta\|^2 + \lambda\|\beta\|_1$$

- The coordinate descent is just soft-thresholding along each coordinate

$$\beta_i = S_{\lambda/\|X_i\|^2}\left(\frac{X_i^T(Y - X_{-i}\beta_{-i})}{X_i^T X_i}\right)$$

- This method works very well on the Lasso problem.

# When Is Coordinate Descent Fast?

- Example: Empirical Risk Minimization

$$\min_{\beta} f(\beta) = \frac{1}{n}\sum_{i=1}^{n}\phi_i(\beta^T X_i) + \lambda\sum_{j=1}^{d}\Omega_j(\beta_j),$$

  where $\phi_i(\beta^T X_i) = L(\beta^T X_i, Y_i)$.
- The derivative at $\beta_j$ is

$$\nabla_j f(\beta) = \frac{1}{n}\sum_{i=1}^{n}\phi_i'(\beta^T X_i)X_{i,j} + \lambda\Omega_j'(\beta_j)$$

  In the first sight, computing $\nabla_j f(\beta)$ needs $O(nd)$ operations.

# When Is Coordinate Descent Fast?

- However, some tricks can help reduce cost of each step. If we store $g_i = \beta^T X_i$, then

$$\nabla_j f(\beta) = \frac{1}{n} \sum_{i=1}^{n} \phi_i'(g_i) X_{i,j} + \lambda \Omega_j'(\beta_j)$$

  and computation cost is $O(n)$.
- If we move $\beta_j^{(k+1)} = \beta_j^{(k)} - t^{(k)} \nabla_j f(\beta^{(k)})$, we can update

$$g_i^{(k+1)} = g_i^{(k)} + X_{i,j} t^{(k)} \nabla_j f(\beta^{(k)}), \qquad i = 1, \ldots, n.$$

  The computation cost is still $O(n)$.
- So the cost of each step is $O(n)$ in coordinate descent, while step cost is $O(nd)$ in gradient descent.

# Block-Coordinate Descent

- Instead of picking up a coordinate, we can select a block of coordinates.

$$x = (x_{(1)}, x_{(2)}, \ldots, x_{(l)}),$$

  where each $x_{(i)}$ is a block.
- In each iteration, we update $i$th block

$$x_{(i)}^{(k+1)} = x_{(i)}^{(k)} - t^{(k)} \nabla_{(i)} f(x^{(k)}).$$

# Example of Block-Coordinate Descent

- Matrix completion problem is

$$\min_{U,V} \sum_{(i,j)\in\mathcal{O}} \left([UV^T]_{i,j} - Y_{i,j}\right)^2$$

- A natural block structure is $U$ and $V$.

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# Componentwise Lipschitz Constant

- Recall $f$ is an $M$-smooth function if

$$\|\nabla f(x) - \nabla f(y)\| \leq M\|x - y\|.$$

- In the analysis of coordinate descent, we need to introduce componentwise Lipschitz constant for the gradient

$$|\nabla_i f(x + \gamma e_i) - \nabla_i f(x)| \leq M_i|\gamma|, \qquad i = 1, \ldots, d.$$

  We can also define their maximum

$$M_{\max} = \max_{i=1,\ldots,d} M_i.$$

- When we compare coordinate descent and gradient descent, we need the following relationship between $M$ and $M_{\max}$

$$M_{\max} \leq M \leq dM_{\max}.$$

# Convergence Analysis: Random Sampling

- We need to introduce another assumption

$$\max_{x \in \mathcal{L}^0} \min_{x^* \in \mathcal{S}} \|x - x^*\| \leq R_0, \tag{1}$$

  where $\mathcal{L}^0 = \{x : f(x) \leq f(x^{(0)})\}$.
- Suppose above assumption in (1) holds, $i^{(k)}$ is randomly chosen from $\{1, \ldots, d\}$ and $t^{(k)} = 1/M_{\max}$. If $f$ is convex, then

$$\mathbb{E}(f(x^{(k)})) - f(x^*) \leq \frac{2dM_{\max}R_0^2}{k}.$$

  If $f$ is $m$-strongly convex, then

$$\mathbb{E}(f(x^{(k)})) - f(x^*) \leq \left(1 - \frac{m}{dM_{\max}}\right)^k \left(f(x^{(0)}) - f(x^*)\right).$$

# Convergence Analysis: Random Sampling

*Proof:* By applying similar arguments in descent lemma, we have

$$\begin{aligned} f(x^{(k+1)}) &= f(x^{(k)} - t^{(k)} \nabla_{i^{(k)}} f(x^{(k)}) e_{i^{(k)}}) \\ &\leq f(x^{(k)}) - t^{(k)} [\nabla_{i^{(k)}} f(x^{(k)})]^2 + \frac{1}{2} (t^{(k)})^2 M_{i^{(k)}} [\nabla_{i^{(k)}} f(x^{(k)})]^2 \\ &\leq f(x^{(k)}) - t^{(k)} [\nabla_{i^{(k)}} f(x^{(k)})]^2 + \frac{1}{2} (t^{(k)})^2 M_{\max} [\nabla_{i^{(k)}} f(x^{(k)})]^2 \\ &\leq f(x^{(k)}) - \frac{1}{2M_{\max}} [\nabla_{i^{(k)}} f(x^{(k)})]^2 \end{aligned}$$

If we take expectation with respect to $i^{(k)}$, we can have

$$\begin{aligned} \mathbb{E}_{i^{(k)}} (f(x^{(k+1)})) &\leq f(x^{(k)}) - \frac{1}{2M_{\max}} \mathbb{E}([\nabla_{i^{(k)}} f(x^{(k)})]^2) \\ &= f(x^{(k)}) - \frac{1}{2dM_{\max}} \|\nabla f(x^{(k)})\|^2 \end{aligned}$$

# Convergence Analysis: Random Sampling

*Proof:* If we define $\phi_k = \mathbb{E}(f(x^{(k)})) - f(x^*)$, then

$$\phi_{k+1} \leq \phi_k - \frac{1}{2dM_{\max}} \mathbb{E}\left(\|\nabla f(x^{(k)})\|^2\right) \leq \phi_k - \frac{1}{2dM_{\max}} \left[\mathbb{E}\left(\|\nabla f(x^{(k)})\|\right)\right]^2.$$

By the convexity of $f$, we have

$$f(x^{(k)}) - f(x^*) \leq \nabla f(x^{(k)})^T (x^{(k)} - x^*) \leq \|\nabla f(x^{(k)})\| \|x^{(k)} - x^*\| \leq R_0 \|\nabla f(x^{(k)})\|.$$

Therefore,

$$\phi_{k+1} \leq \phi_k - \frac{1}{2dM_{\max}R_0^2} \phi_k^2.$$

Equivalently, we have

$$\frac{1}{\phi_{k+1}} \geq \frac{1}{\phi_k} + \frac{1}{2dM_{\max}R_0^2} \frac{\phi_k}{\phi_{k+1}} \geq \frac{1}{\phi_k} + \frac{1}{2dM_{\max}R_0^2}.$$

*Proof:* So we can conclude when $f$ is convex, then

$$\frac{1}{\phi_k} \geq \frac{k}{2dM_{\max}R_0^2}.$$

If $f$ is $m$-strongly convex, we know

$$\|\nabla f(x^{(k)})\|^2 \geq 2m(f(x^{(k)}) - f(x^*)).$$

Hence,

$$\phi_{k+1} \leq \phi_k - \frac{m}{dM_{\max}}\phi_k = \left(1 - \frac{m}{dM_{\max}}\right)\phi_k.$$

# Convergence Analysis: Cyclic

- Suppose above assumption in (1) holds, $i^{(k)} = (k \mod d) + 1$ and $t^{(k)} = t \leq 1/M_{\max}$. If $f$ is convex, then

$$f(x^{(k)}) - f(x^*) \leq \frac{(4d/t)(1 + dM^2t^2)R_0^2}{k+8}.$$

  If $f$ is $m$-strongly convex, then

$$f(x^{(k)}) - f(x^*) \leq \left(1 - \frac{m}{(2/t)(1 + dM^2t^2)}\right)^{k/d} \left(f(x^{(0)}) - f(x^*)\right).$$

  Here $k = d, 2d, 3d, \ldots$.

- The above bound suggests cyclic coordinated descent doesn't have advantage over gradient descent.

# Convergence Analysis: Random Permutation

- The convergence rate in random permutation is similar with the one in cyclic (actually we don't event modify the proof).
- However, the empirical performance suggests random sampling is similar to random permutation. So we still need some new proof and results.

# Coordinate Descent

- Coordinate descent method is a descent method, i.e.

$$f(x^{(k+1)}) \leq f(x^{(k)})$$

- Coordinate descent method converge to the minimum, when $f$ is strictly convex and smooth.
- Coordinate descent method might not converge if $f$ is non-convex or non-smooth. It can stuck at some non-differentiable point.
- The coordinate descent method is widely used in statistics and machine learning fields.